

main.py

Visualize

Run

```
1 #MOHAMMED SHAHID S
2 class RuleBasedSystem:
3     def __init__(self, facts, rules):
4         self.facts = set(facts)
5         self.rules = rules
6
7     def forward_chain(self):
8         iterations = 0
9
10        while True:
11            new_fact_added = False
12
13            for rule in self.rules:
14                # Check if all conditions of the rule exist in current facts
15                if all(cond in self.facts for cond in rule['if']):
16                    if rule['then'] not in self.facts:
17                        print(
18                            f"Rule Triggered: IF {rule['if']} "
19                            f"THEN Add {rule['then']}"
20                        )
21
22                        self.facts.add(rule['then'])
23                        new_fact_added = True
24
25            if not new_fact_added:
```

Output

```
Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_n
Final Knowledge Base Facts: {'All_humans_are_mortal', 'Socrates_is_mortal',
'Socrates_is_human'}
```

```
=== Code Execution Successful ===
```